# Ajinkya Vivek Nagarkar

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Los Angeles, California

## Education

**University of Southern California** — Aug 2025 – May 2027
*Master of Science in Computer Science* — Los Angeles, CA

**Dr. Vishwanath Karad MIT World Peace University** — Jul 2019 – May 2023
*Bachelor of Technology (Computer Science and Engineering)* — Pune, India
CGPA: 9.56/10

## Professional Experience

**Capgemini Technology Services India** — Dec 2024 – Jul 2025
*Analyst* — Mumbai, India

- Engineered a data ingestion pipeline processing 100K+ sensor records for predictive maintenance modeling.
- Performed EDA and feature engineering on structured sensor data to build supervised learning models.
- Trained and evaluated regression models achieving $R^2$ up to 0.93 on validation datasets.
- Embedded predictive outputs into operational dashboards, enabling realtime equipment monitoring.

**Asha Electronics** — Jul 2023 – Jun 2024
*IoT Application Developer Intern* — Pune, India

- Developed a full-stack IoT monitoring platform using Angular and Node.js for real-time distributed sensor data visualization.
- Implemented offline-first sync and Firebase-backed auth, ensuring resilience under intermittent connectivity.

## Projects

**Evidence-Based RAG for Multi-Hop Q&A**

- Built a hallucination-resistant multi-hop QA system combining hybrid BM25+FAISS retrieval, fuzzy title matching, and cross-encoder reranking across 7,405 HotpotQA examples.
- Improved Joint F1 from 19.6% to 43.2% zero-shot; achieved BERTScore F1 of 94.3% versus 46.8% EM exposing limitations on generative answers.
- Implemented NLI-based verification and citation selection to anchor responses in retrieved evidence, eliminating hallucinated references.

**Fungi Image Classification via Transfer Learning**

- Systematically evaluated five pretrained CNN architectures (ResNet50, ResNet101, EfficientNetB0, VGG16, DenseNet201) as feature extractors for 5-class fungi image classification.
- Applied aggressive augmentation (crop, zoom, rotation, flip) with L2 regularization, batch normalization, and dropout with early stopping over 100 epochs.
- ResNet101 topped test accuracy at 69.2% (F1: 66.4%); DenseNet201 led on AUC at 90.0%, with all architectures exceeding 85% AUC.

**Solar Power Plant Maintenance Prediction**

- Constructed an end-to-end regression framework on photovoltaic sensor and meteorological data to forecast solar panel degradation.
- Trained Random Forest, LightGBM, and Linear Regression models with K-fold cross-validation, improving prediction accuracy from 67% to 82.9%.

## Patents and Publications

**Engaging Attackers with a Highly Interactive Honeypot System Using LLM-based Interaction Modeling**
*IEEE-Xplore, ICCUBEA-23*
Developed a ChatGPT powered honeypot that simulates vulnerable systems to deceive attackers and profile their tactics. The system dynamically responds to attacker commands, enabling collection of threat intelligence on adversarial behavior patterns.

**A Crypto Wallet for Farmers**
*Indian Patent Office*
Co-invented "Apna Wallet", a Raspberry Pi-based hardware crypto wallet with fingerprint biometrics and CBDC support, enabling direct government subsidy disbursement to digitally illiterate rural farmers.

## Skills

**Machine Learning and AI**
RAG, Transformers, LLMs, Hyperparameter Tuning, Cross-Validation

**Data & Analytics**
Feature Engineering, EDA, Model Evaluation, Data Visualization

**ML Libraries**
Scikit-learn, PyTorch, Pandas, NumPy, Matplotlib, Seaborn, FAISS

**Programming Languages**
Python, C++, SQL, TypeScript, JavaScript

**Frameworks**
React, Angular, Node.js, REST APIs

**Tools & Platforms**
Git, Docker, Linux, GitHub Actions.